

AKSHAY BORSE

Auburn, WA | 98092 | +1-425-336-9852 | amsborse@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Senior Software Engineer with 6+ years of experience architecting large-scale distributed systems, leading AI/ML integrations, and delivering enterprise-grade cloud solutions across Amazon and Microsoft. Specializes in high-throughput microservices, applied machine learning, intelligent automation, and systems that power mission-critical business operations.

Proven track record of modernizing architectures, elevating operational excellence, and driving measurable business outcomes, including multi-million-dollar annual profit gains, 25% cost reductions, 16% improvement in delivery accuracy, and 250k+/month in cloud-cost optimizations.

Recognized for strong cross-org leadership, deep ownership, and the ability to simplify complex systems. Adept at aligning engineering with business strategy, mentoring teams, and building long-term technical foundations while shipping at high velocity. Beyond engineering, I explore how technology, culture, and AI shape the human experience, and I write about creativity, growth, and meaning in the age of intelligent systems.

SKILLS

- **Languages:** Java, Python, C#, JavaScript, TypeScript
- **Backend:** Microservices, REST, gRPC, Event-Driven Systems, Serverless
- **AI/ML:** SageMaker, Bedrock, LLM Integration, RAG, XGBoost, ML Ops
- **Infra/Tools:** Docker, App Mesh, OpenSearch, Kibana, Kusto, CloudWatch, Bicep, ARM
- **Cloud:** AWS (DynamoDB, Lambda, ECS, S3, Kinesis, Step Functions), Azure (Functions, Cosmos DB, Event Hub)
- **Other:** Architecture Design, Observability, CI/CD, On-Call Leadership, Cost Optimization

EXPERIENCE

Senior Software Engineer

09/2025 to Current

Microsoft

Redmond, WA

(C#, .NET, Azure Functions, Cosmos DB, ARM/Bicep, Agentic AI, Purview, Event Hub, Kusto)

- Developing agentic risk detection capabilities in Microsoft Purview to identify anomalous or high-risk agent behavior across digital and embodied agents, strengthening data governance and insider-threat detection.
- Built adaptive agent-protection models enabling tenant-configurable risk scoring and dynamic policy enforcement, improving governance precision for enterprise customers.
- Enhancing large-scale telemetry and audit signal ingestion using Event Hub + Cosmos DB for near-real-time compliance analytics.
- Partnering cross-org with Azure security and Purview governance teams to shape the next generation of AI-centric security policy frameworks.

Software Engineer II

08/2019 to 08/2025

Amazon

Seattle, WA

(Java, Python, gRPC, REST, AWS Lambda, DynamoDB, ECS, Step Functions, Redis, Kinesis, OpenSearch, SageMaker, Bedrock, CI/CD)

Architecture & Platform Initiatives

- Built a microservice bootstrap framework that reduced multi-week setup to hours, saving 40–50 weeks of engineering time annually across orgs.
- Enhanced the platform to support multi-service deployment within a single pipeline, streamlining operations for 60+ distributed services.
- Optimized Docker pipelines, cutting image size by 83% and build times by 50%, saving 60GB+ in storage and accelerating CI/CD speed across teams.
- Led org-wide logging and metrics modernization, cutting CloudWatch spend by \$250K+/month while improving signal quality and debugging velocity.

High-Impact Systems & Services

- Built a Restock Alerting Service that drove a 13% sales lift for previously cancelled ASINs.
- Built the Just-In-Stock real-time inventory engine on native AWS, improving early delivery rates by 16% and enabling dynamic fulfillment optimization.
- Delivered a self-service delivery date modification UI + backend for Amazon Business customers, removing 2–3 hrs of manual ops per request.

- Designed a Replay Scheduler leveraging Step Functions to orchestrate delayed event replays (up to 1 year), eliminating cron-based failures and improving resilience.

AI/ML, Predictive Systems & LLM Integrations

- Led architecture for a RAG-based intelligence service using SageMaker-hosted embeddings + vector search, improving delivery prediction accuracy and driving 7% global sales lift.
- Integrated an XGBoost-based delay-risk classifier, enabling early SLA breach predictions and improving customer trust through proactive alerts.
- Designed and launched backend infrastructure for a quantile regression model powering delivery-range predictions, yielding ~\$28M annualized profit.
- Modernized 37 customer-facing email templates using Bedrock-backed LLM personalization, ensuring compliance while improving message quality and brand consistency.

Distributed Systems, Scale, and Operations

- Architected a real-time event orchestration layer supporting 100K+ TPS across 16 microservices, balancing REST and gRPC protocols.
- Led cross-org coordination for Prime Day and Black Friday load testing across 60+ critical services, proactively identifying scaling gaps.
- Built a dry-run simulation framework enabling high-visibility scenario analysis for non-engineering stakeholders, reducing deep-dive cycles from hours to minutes.
- Drove cultural improvements: on-call deep dives, reusable design docs, post-mortems, automated wiki tooling, cutting Sev2 alerts by 60% and accelerating onboarding.

Software Engineer

12/2018 to 01/2019

Liquiron

San Jose, CA

- Re-architected authentication via OAuth 2.0 + Passport.js, improving modularity and security.
- Migrated backend services to Firebase for fully serverless operation, eliminating local server overhead.
- Implemented 4 REST APIs powering 10 front-end SPA features, reducing server load by 36%.

Software Engineer

09/2016 to 06/2017

Persistent Systems

Pune, IN

(CNN, EC2, NodeJS, Python, Quasar)

- Built a real-time CNN sentiment classifier achieving 87% accuracy across video and image inputs.
- Converted legacy systems into a single-page application, improving response times by 25%.
- Improved offline resilience via caching strategies, enhancing responses by 9% on unreliable networks.

EDUCATION

M.S., Computer Science & Engineering

06/2019

Santa Clara University

Santa Clara, CA

B.E., Computer Engineering

06/2016

Pune Institute of Computer Technology

Dhankawadi, IN

PROJECTS

<https://github.com/ambrose>

- Hero Recommendation System for Dota 2 - Trained on 50k matches; engineered full ingestion pipeline using the Dota API.
- Natural Language to SQL Query Generator - Schema-independent NLP-to-SQL engine; achieved 85% accuracy on a 10k test set.
- Face & Motion Recognition System - PCA + L2-norm face recognition (99% accuracy); CNN + edge detection for motion (~77% accuracy).

ACHIEVEMENTS

- Author of essays on creativity, identity, and navigating the AI age with depth and originality.
- Awarded Just Do It Award for building automated microservice deployment tooling at Amazon.
- Winner of Most Innovative Project at the Bronco Hackathon, Santa Clara University